EisaX Intelligence Platform

INVESTMENT INTELLIGENCE · INSTITUTIONAL GRADE

EQUITY

Date: April 15, 2026
Sector: Exposure
Source: EisaX AI + TradingView

Platform: **EisaX** Region: **GCC / MENA** Type: **Institutional Equity Analysis**

Classification: **CONFIDENTIAL — Professional Investors Only**

✅ Portfolio uploaded: **NVDA, AAPL, MSFT, TSLA, CASH**

📊 EisaX Portfolio Risk Report

Date: April 15, 2026 | **Period:** 1 Year | **Risk-Free Rate:** 4.5% (US T-Bill)

🎯 Executive Summary

> Equity sleeve returned **+47.6%**; estimated total-portfolio return (including 15.0% cash) is **+40.4%**. Vs S&P 500 **+30.0%**: equity alpha **+17.5%**, estimated total alpha **+10.4%**. Risk profile (total beta): **Aggressive** 🔴 (1.68). Sharpe (equity sleeve): **~1.6**. Strategic guidance: Strong performance, but concentration-driven with elevated risk — consider gradual de-risking and defensive diversification.

📋 Holdings

Ticker	Weight (Total)	Weight (Equity Sleeve)	Sector	Beta	P/E	Div Yield
NVDA	30.0%	35.3%	Technology	2.33	40.5	0.02%
AAPL	20.0%	23.5%	Technology	1.11	33.7	0.40%
MSFT	20.0%	23.5%	Technology	1.11	25.7	0.89%
TSLA	15.0%	17.6%	Consumer Cyclical	1.92	361.0	0.00%
CASH	15.0%	—	—	0	—	0%

> **Weighted Dividend Yield (Total Portfolio):** 0.26% | **Equity Sleeve:** 0.31%

📈 Risk Metrics

Method: Historical Simulation (252 trading days) · rf = 4.5% · equity-sleeve normalized

Metric	Value	Assessment
1Y Return (Equity Sleeve)	+47.6%	🟢 Strong
Estimated 1Y Return (Total Portfolio)	+40.4%	Includes 15.0% cash drag
vs S&P 500 (Alpha)	+17.5%	🟢 Outperforming benchmark
Annualized Volatility	23.8%	🟡 Moderate
Sharpe Ratio (1Y, rf=4.5%)	1.58	🟢 Excellent
Sortino Ratio (1Y, rf=4.5%)	2.48	🟢 Good downside-adjusted
Portfolio Beta (Total Weight)	1.68	🔴 High Risk
Portfolio Beta (Equity Sleeve)	1.68	Normalized over non-cash assets
VaR 95% 1-Day (Historical)	-2.34%	95% of days, loss ≤ this
CVaR 95% (Expected Shortfall)	-3.03%	Avg loss **when** VaR is breached — tail risk
Max Drawdown (1Y)	-20.7%	Worst peak-to-trough
Max Drawdown Duration	114d	Longest time underwater
Portfolio Div Yield (Total)	0.26%	Weighted annual income
Portfolio Div Yield (Equity Sleeve)	0.31%	Normalized over non-cash assets
Rolling Sharpe (63d)	—	Current (last 63d): **-0.34**

🏭 Sector Exposure

Sector	Weight	HHI Contribution
Technology	82.4%	0.678
Consumer Cyclical	17.6%	0.031

Sector HHI: 0.709 — 🔴 Highly Concentrated *(0=perfect diversification, 1=single sector)*

🔗 Correlation & Diversification

	NVDA	AAPL	MSFT	TSLA
NVDA	1.00	0.31	0.44	0.43
AAPL	0.31	1.00	0.19	0.34
MSFT	0.44	0.19	1.00	0.30
TSLA	0.43	0.34	0.30	1.00

Metric	Value	Interpretation
Avg Pairwise Correlation	0.33	🟢 Low co-movement between holdings
Effective N (eigenvalue)	2.9	🟢 Good — true independent bets in portfolio

⚠️ **Note:** Even without pairs >0.70, Effective N = 2.9 indicates portfolio behaves like **2.9 independent assets** — not 4. Sector concentration (not just pairwise correlation) is the real risk driver here.

🧪 Stress Testing

Historical crises use actual observed returns + sector-adjusted amplification (NOT linear beta)

Scenario	SPX Actual	Portfolio Est.	Notes
🔴 2022 Rate Shock (actual)	-19.5%	**-28.3%**	*Tech -33% vs SPX -19.5% in 2022*
🔴 COVID Crash Mar 2020	-34.0%	**-40.8%**	*SPX -34% in 33 days*
🔴 2008 GFC (Sep–Nov)	-42.0%	**-37.8%**	*Financials led; tech -40%*
🔴 Bear Market -30% Scenario	-30.0%	**-39.0%**	*Projected; tech typically amplified*
🟢 2023 AI Bull Run (actual)	+26.5%	**+47.7%**	*Tech +55% vs SPX +26.5%*
🟢 Rate Cut Cycle +15%	+15.0%	**+21.0%**	*Growth stocks benefit most*

Tech Weight: 100% of equity. In risk-off events, tech typically amplifies market moves by 1.2–1.8×. Linear beta alone (equity β=1.68) understates true drawdown risk.

💡 EisaX Risk Assessment

🔴 **Aggressive** — total β=1.68 (equity β=1.68), CVaR=-3.03%/day, Eff.N=2.9 bets. 82% in Technology. In a 2022-style selloff your equity sleeve would have lost ~33% vs SPX -19.5%. Diversification is urgently needed.

📐 Performance Attribution (1Y, Equity Sleeve)

Brinson-Hood-Beebower: each holding's contribution to equity-sleeve return

Ticker	Weight (Equity Sleeve)	1Y Return	Contribution	Attribution
NVDA	35.3%	+76.8%	+27.09pp	🟢
TSLA	17.6%	+54.8%	+9.68pp	🟢
AAPL	23.5%	+32.0%	+7.53pp	🟢
MSFT	23.5%	+7.6%	+1.79pp	🟢
Residual (model)	—	—	+1.47pp	⚪
TOTAL	100%	—	**+47.56pp**	

🏆 **Alpha driver:** NVDA contributed +27.1pp (+76.8% × 35% weight). S&P 500 returned +30.0% — alpha gap of +17.5pp.

🧬 Factor Exposure Analysis

Fama-French proxy: factor tilts computed from fundamentals + price momentum

Factor	Portfolio Exposure	Interpretation
Market (Beta)	**+1.68**	🔴 High market sensitivity
Growth (P/E tilt)	**+0.50**	🔴 Strong growth tilt — high valuation risk
Size (SMB proxy)	**-0.84**	🟢 Large-cap dominated (low SMB)
Momentum (12M)	**+0.46**	🟢 Strong momentum

Factor exposures are approximated from fundamentals. For precise Fama-French loadings, a 3-year return regression against HML/SMB/MKT factors is required.

⚙️ Institutional Optimization Engine

Convex QP · CLARABEL solver · Zero-tolerance constraint enforcement

🔍 Constraint Diagnostics

Reference: $100,000 portfolio · normalized weights

Constraint	Limit	Current	Status	Required Fix
Single stock: **NVDA**	≤20%	35.3%	🔴 −15.3pp	Sell ~$15,294 of NVDA
Single stock: **AAPL**	≤20%	23.5%	🔴 −3.5pp	Sell ~$3,529 of AAPL
Single stock: **MSFT**	≤20%	23.5%	🔴 −3.5pp	Sell ~$3,529 of MSFT
Sector: **Technology**	≤40%	82.4%	🔴 −42.4pp	Reduce by ~$42,353 / add non-Technology
Portfolio beta	≤1.30	1.68	🔴 +0.38	Add low-β assets: TLT, GLD, BRK-B, XLV

💸 **Estimated rebalancing:** ~$22,353 turnover (22.4% of portfolio) to reach basic mandate compliance

📊 Mandate Mode Feasibility

	🏛️ Conservative	⚖️ Balanced	🚀 Aggressive
Max single stock	10%	15%	20%
Max portfolio β	≤1.00	≤1.20	≤1.50
Max sector weight	30%	35%	40%
Single-stock achievable	❌ Need 10+ holdings	❌ Need 7+ holdings	❌ Need 5+ holdings
Beta achievable	❌ Min β=1.13	✅	✅
Sector cap achievable	⚠️ Current assets only	⚠️ Current assets only	⚠️ Current assets only

⚠️ **Constraint violations — current holdings cannot satisfy any mandate:**

- 4 holdings × 10% = 40% < 100% (need ≥10 holdings for this mandate)
- Min achievable β = 1.13 exceeds cap β≤1.00

💡 **To unlock full institutional compliance, add any of:**

- Add 1+ new assets OR use Aggressive mandate (4×20%→80% min)
- Add 3+ new assets OR use Aggressive mandate (4×20%→80% min)
- Add 6+ new assets OR use Aggressive mandate (4×20%→80% min)
- Add diversifying sectors: XLV (Healthcare), TLT (Bonds), GLD (Gold), BRK-B, XLP
- Add: TLT (β≈−0.5), GLD (β≈0.1), USMV (β≈0.7), BRK-B (β≈0.9)

📊 Optimal Weight Comparison — Strict Institutional Mode

🔒 **Strict Mode:** Only fully-compliant solutions shown. ❌ `INFEASIBLE` = mandate cannot be achieved with current holdings — see asset suggestions above.

Ticker	Current	🏛️ Conservative	⚖️ Balanced	🚀 Aggressive
NVDA	35.3%	❌ N/A	❌ N/A	❌ N/A
AAPL	23.5%	❌ N/A	❌ N/A	❌ N/A
MSFT	23.5%	❌ N/A	❌ N/A	❌ N/A
TSLA	17.6%	❌ N/A	❌ N/A	❌ N/A
--------	---------	------	------	------
Sharpe	1.58	❌	❌	❌
Beta	1.68	❌	❌	❌
CVaR/day	-3.03%	❌	❌	❌
Ann. Vol	23.8%	❌	❌	❌
Compliance	Baseline	❌ INFEASIBLE	❌ INFEASIBLE	❌ INFEASIBLE

🔍 Exploratory Allocation — Pre-Asset Expansion

> 🔍 **Exploratory Allocation (Non-Mandate Scenario)** — Shows the best achievable structure with current holdings before adding diversifying assets. Constraints adjusted where mathematically required. This is a planning scenario, not a mandate-compliant allocation.

Ticker	Current	🔍 Exploratory
NVDA	35.3%	16.5%
AAPL	23.5%	30.0%
MSFT	23.5%	30.0%
TSLA	17.6%	23.5%
--------	---------	----------------
Sharpe	1.58	1.38
Beta	1.68	1.50
CVaR/day	-3.03%	-2.98%
Compliance	❌ Breach	🔍 Exploratory — Pre-Expansion

> 🔍 🔍 *EXPLORATORY MODE — constraints adjusted: single-stock cap adjusted to 30% (mandate 20% infeasible with 4 holdings); sector cap lifted (was 40%, current holdings prevent compliance)*

📋 Execution Plan → 🔍 *Exploratory Allocation (Non-Mandate Scenario)*

Pre-asset expansion — exploratory scenario

🔍 **Exploratory Allocation** — These trades show the optimal structure achievable with current holdings before adding new assets. This is a planning scenario to guide asset expansion decisions — **not a mandate-compliant allocation.**

Action	Ticker	From	To	Δ	$ on $100k	Reason
🔴 **SELL** ↓	NVDA	35.3%	16.5%	-18.8pp	$18,818	Exceeds 20% mandate cap
🟢 **BUY** ↑	AAPL	23.5%	30.0%	+6.5pp	$6,471	Underweight vs risk-optimal
🟢 **BUY** ↑	MSFT	23.5%	30.0%	+6.5pp	$6,471	Underweight vs risk-optimal
🟢 **BUY** ↑	TSLA	17.6%	23.5%	+5.9pp	$5,877	Underweight vs risk-optimal

📐 Multi-Objective Trade-off Analysis

Strategy	Sharpe	CVaR/day	Beta	Ann.Vol	Status
Current portfolio	1.58	-3.03%	1.68	23.8%	Baseline
🏛️ Conservative	—	—	—	—	❌ INFEASIBLE — single-stock cap 10% requires ≥10 holdings (current: 4); beta cap β≤1.00 unachievable (min possible β=1.13); sector cap 30% unachievable: 'Technology' has 3/4 holdings
⚖️ Balanced	—	—	—	—	❌ INFEASIBLE — single-stock cap 15% requires ≥7 holdings (current: 4); sector cap 35% unachievable: 'Technology' has 3/4 holdings
🚀 Aggressive	—	—	—	—	❌ INFEASIBLE — single-stock cap 20% requires ≥5 holdings (current: 4); sector cap 40% unachievable: 'Technology' has 3/4 holdings

🔮 Post-Rebalance Simulation

Best mandate: 🔍 Exploratory Allocation (Non-Mandate Scenario) · Pre-asset expansion — exploratory scenario

Metric	Before	After	Δ	Verdict
Sharpe Ratio	1.58	1.38	-0.20	🟡 Similar
Ann. Volatility	23.8%	22.9%	-0.92	🟡 Similar
Portfolio Beta	1.68	1.50	-0.18	✅ Reduced
CVaR 95%/day	-3.03%	-2.98%	+0.06	🟡 Similar
Single-stock cap	🔴 Breach	🔍 Exploratory	—	🔍 Exploratory scenario
Beta mandate	🔴 Breach	✅ OK	—	🔍 Exploratory

🧠 CIO Deep Analysis (AI-Powered)

1. 📋 Executive Assessment

The portfolio has generated significant absolute (+40.4%) and risk-adjusted (Sharpe 1.58) returns, outperforming the SPX by 17.5%. However, the alpha quality is questionable given the extreme concentration in high-beta Technology (82% sector weight) and a total portfolio beta of 1.68. The Sharpe ratio is elevated but supported by a strong bull market; the recent 63-day trend is declining (-0.34). Performance is highly conditional on market direction, as evidenced by stress tests where losses exceeded the SPX in most bear scenarios.

2. 🚨 Top 3 Risks

- **Critical - Sector Concentration:** 82% Technology exposure creates systemic risk. A sector downturn could trigger the -39.0% equity sleeve loss shown in the -30% SPX scenario.
- **High - Leveraged Beta (1.68):** Magnifies losses in market corrections. The 2022 Rate Shock stress test shows a -28.3% equity loss vs. SPX -19.5%. Trigger: A sustained rise in real yields or growth scare.
- **Medium - Single-Stock Concentration in NVDA:** At 35% of equity, idiosyncratic risk is elevated. A fundamental disappointment could trigger a loss exceeding its individual contribution to the daily CVaR of -3.03%.

3. 🔄 Rebalancing Plan

Target Total Weights: NVDA 22.5%, AAPL 17.5%, MSFT 17.5%, TSLA 12.5%, Cash 30.0%.
Rationale: Reduces single-stock (NVDA) and sector concentration, lowering tail risk (CVaR). Increases cash to 30% to lower portfolio beta and provide dry powder. The equity sleeve's Effective N (2.9) remains similar, but higher cash reduces total portfolio volatility.

4. ➕ Suggested Additions

- **XLE (Energy ETF):** Beta ~0.9, expected Sharpe ~0.7. Low/negative correlation to Tech would improve Effective N and sector HHI.
- **JPM (Financials):** Beta ~1.3, expected Sharpe ~1.0. Provides cyclical exposure less tied to tech multiples, reducing avg. pairwise correlation.
- **ED (Utilities):** Beta ~0.3, expected Sharpe ~0.5. Defensive, low-correlation addition to lower portfolio beta and volatility.

5. 📊 Tax & Income Note

Substantial unrealized capital gains are likely, particularly in NVDA (+47.6% sleeve return). A full rebalance would realize significant tax liability. The portfolio has a severe income gap: 0.26% total dividend yield vs. a 4.5% risk-free rate, creating a ~4.24% yield shortfall versus cash.

6. ✅ EisaX Final Verdict

Rating: **Aggressive**.
No-Action Case: Portfolio remains a high-beta, concentrated tech bet, likely outperforming strongly in bull markets but vulnerable to severe drawdowns.
Alternative Scenario: A phased rebalance, trimming NVDA and TSLA on strength into XLE and JPM, while allowing cash to drift toward 25%.
Advisory Suggested Action: Implement the rebalancing plan in a tax-aware manner, using lots with lowest cost basis last.

Decision Boundary: This analysis provides strategic guidance, not execution instructions.

🎯 Decision Discipline Layer

- **Confidence:** 70%
- **Risk Posture:** Aggressive
- **Primary Uncertainty:** concentration regime sensitivity and rolling Sharpe currently -0.34
- **No-Action Case:** If risk metrics stay stable and no catalyst break occurs before the next quarterly review, HOLD/no-trade remains a valid decision.
- **Alternative Scenario:** If AI mega-cap momentum persists and earnings revisions remain positive, concentrated tech could continue to outperform in the near term.

> **Decision Boundary:** This analysis provides strategic guidance, not execution instructions.

To analyze any individual stock: type `analyze TICKER`

📋 Audit Trail

For compliance, reproducibility, and institutional trust

Field	Value
Report ID	`977c6451-fa64-4d4a-896e-0a988583802e`
Generated	2026-04-15 19:42:18 UTC
Price Data As-Of	2026-04-15
Period Analysed	2025-04-15 → 2026-04-15 (252 trading days)
Data Source	Yahoo Finance (yfinance) — 15-min delayed
Tickers Fetched	NVDA, AAPL, MSFT, TSLA
Risk-Free Rate	4.50% (US 3-Month T-Bill)
Methodology	Historical Simulation · Brinson Attribution · Fama-French proxy
Optimizer	CLARABEL convex QP (cvxpy) — strict constraint enforcement
Report Hash	`sha256:8ecc54bacb8e22a6...`

> 🔁 **Reproduce this report:** Save snapshot ID `977c6451-fa64-4d4a-896e-0a988583802e` → call `GET /v1/portfolio/snapshot/977c6451-fa64-4d4a-896e-0a988583802e`